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The Standard Model

The Theory of Fundamental Particles

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Introduction

Every fundamental particle known to science, and three of the four forces that act between them, fits on a diagram small enough to print on a coffee mug. Six quarks, six leptons, four force carriers, and one Higgs boson, organized by charge, spin, and mass, with a handful of rules binding them together: that is the Standard Model of particle physics, the framework that took a century of experiment and a half-century of theory to assemble, and that now passes every test the world's largest machines can devise. It is, by a wide margin, the most successful scientific theory ever constructed. It is also, by its authors' own repeated testimony, incomplete, and the tension between those two facts is the story this book tells.

The Standard Model grew from the wreckage of the particle zoo. In the 1950s and 1960s, accelerators were discovering new particles faster than theory could name them, and the physics of the smallest scales looked for a while like botany, a catalog of species with no underlying principle. The rescue came from an unexpected direction: symmetries, the mathematical study of what stays the same when things change, and gauge fields, the machinery that turns a symmetry into a force. Piece by piece, the zoo was tamed. The dozens of hadrons were revealed as arrangements of a handful of quarks. The weak force, the agent of radioactive decay, was married to electromagnetism in a single electroweak theory. The strong force that binds atomic nuclei was rewritten as a gauge theory of color charge, and the whole assembly was shown to be mathematically consistent, calculable to arbitrary precision, by a single proof of renormalizability.

This book follows that assembly in the order the physics demands. The opening chapters tell the discovery history: Dirac's equation and the antimatter it forced into existence, Yang and Mills's extension of gauge theory beyond electromagnetism, the quark hypothesis and the color charge it required, Glashow, Weinberg, and Salam's electroweak unification, and the Higgs mechanism that gave the whole structure its mass. The middle chapters explain the model itself: what a gauge theory is, how the three interactions work, what the elementary particles are and how they are classified, and how the mathematics of symmetry groups, Lagrangians, and quantum fields carries it all. Later chapters turn to experiment: the colliders that confirmed the model's predictions one by one, the W and Z bosons, the top quark, and finally the Higgs boson in 2012, the last missing piece, found forty-eight years after it was proposed.

The final chapters take up what the model does not explain, because that is where the frontier now lives. The Standard Model describes three of the four fundamental interactions; gravity stands outside it, unmerged and unmerged-able by any current method. It cannot account for the dark matter that outweighs ordinary matter five to one, the dark energy that accelerates the cosmos, or the matter-antimatter asymmetry that lets anything exist at all. It has no explanation for why there are three generations of particles, why the masses take the values they do, or why over a dozen of its own parameters must be measured rather than derived. The model's failures are as informative as its successes, and they chart the course of the physics that comes next. The story begins with a theory that worked too well, an equation whose implications its own author did not believe.

A word on how this book is organized. It follows the Aira Astralis Standalone format, and the path runs from history to machinery to frontier. The early chapters tell how the model was discovered, era by era, and then explain what it says: the census of particles, the ledger of charges, the three forces and their carriers. The middle chapters turn to the mathematics and the experiments, the two halves of the model's authority, showing how symmetry generates force, how the Lagrangian is quantized, and how the colliders turned each prediction into an observation. The later chapters place the model in its contexts, from the atom to the cosmos; survey the research programs that now audit its every decimal; count the discoveries that built it; and close with its challenges, its future, its trivia, and its meaning. No equations appear in these pages, and none are needed: the model's ideas are fully tellable in prose, and the prose keeps faith with the physics throughout.

The scale of the model's success deserves its own framing before the story begins. When the tau neutrino's discovery closed the matter census in 2000, and the Higgs boson's discovery closed the full census in 2012, the Standard Model became the first physical theory in history with every one of its predicted constituents observed. Its precision is of the same order: the electron's magnetic moment, the model's electromagnetic sector's sharpest observable, matches theory and experiment through more than a dozen significant figures, and no rival candidate discrepancy has survived audit anywhere in the model's tested range. Yet the model describes only a twentieth of the universe's energy content, ignores the one force everyone can feel, and leaves its own basic numbers unexplained. Both halves of that sentence are equally true, and the chapters ahead give each its due: the model as the most complete description of matter ever achieved, and the model as the precisely drawn boundary of that

achievement, the boundary where the physics of the coming century begins.

Chapter 1: The Discovery of The Standard Model

The discovery of the Standard Model is not one discovery but a chain, and the chain's first link was forged in 1928, when Paul Dirac combined quantum mechanics with special relativity in a single wave equation for the electron. The Dirac equation explained the electron's spin and magnetic moment as consequences of relativity, and it carried an implication Dirac initially refused: the equation demanded a particle of the same mass and opposite charge, a positive electron. When the positron turned up in cosmic-ray tracks in 1932, antimatter passed from embarrassment to prediction, and the lesson was absorbed into the physics of the century: a good equation knows more than its author. The Dirac equation's success established the pattern the Standard Model would follow, of particles derived from symmetry rather than observed and cataloged.

The next link came in 1954, when Yang Chen-Ning and Robert Mills asked a question with enormous consequences: what happens if gauge theory, the framework that had triumphed in electrodynamics, is generalized from the simple symmetry of electric charge to the more complicated non-Abelian symmetries, where the order of transformations matters? Their answer was a family of field theories in which the force carriers themselves carry charge and interact with one another. At the time, the Yang-Mills theory looked like a solution in search of a problem; within twenty years, it was the backbone of everything, and the strong and weak forces are both, in modern language, Yang-Mills gauge theories. The theorists of the 1950s also delivered the first crack in the old physics: in 1957, Chien-Shiung Wu's experiment on cobalt decay showed that parity, the mirror symmetry physicists had assumed was fundamental, was violated by the weak interaction, whose charged carriers couple only to left-handed particles. The weak force was revealed as odder than any classical intuition allowed.

The 1960s supplied the matter. In 1961, Sheldon Glashow proposed the structure that would unify electromagnetism and the weak interaction into a single electroweak theory, with the photon joined by three heavy bosons, the W+, W-minus, and Z. In 1964, Murray Gell-Mann and George Zweig independently proposed the quark model, taming the particle zoo by showing its hundreds of hadrons were arrangements of three fundamental constituents. The same year, Oscar Greenberg implicitly introduced the color charge of quarks, the three-valued property that would become the charge of the strong force; and, also in 1964, three groups proposed the mechanism by which the electroweak bosons acquire mass, with the boson of Peter Higgs's name attached to it. The pieces were now on the table: a symmetry group for the weak force, a constituent model for matter, a charge for the strong force, and a mechanism for mass. What held them together was not yet known.

The synthesis years followed. In 1967, Steven Weinberg and Abdus Salam independently incorporated the Higgs mechanism into Glashow's electroweak structure, giving the unified theory its modern form, with the photon emerging massless and the W and Z heavy, from a single broken symmetry. In 1970, Glashow, John Iliopoulos, and Luciano Maiani introduced the GIM mechanism, which suppressed the flavor-changing processes the theory otherwise predicted; its side effect was the prediction of a fourth quark, charm, found in 1974. In 1971, Gerard 't Hooft, a student in Utrecht, completed the theoretical edifice by proving that Yang-Mills gauge theories, including the electroweak model, are renormalizable, that their infinities can be tamed and their predictions computed to arbitrary precision. That proof converted the electroweak theory from a speculative structure into a calculable science, and its significance was recognized with a Nobel Prize within a few years.

The strong force was brought into the fold in 1973, when David Gross, Frank Wilczek, and David Politzer discovered that non-Abelian gauge theories have asymptotic freedom, the property that the strong force weakens at short distances. Their finding explained why quarks behave as free particles when probed hard, and why they can never be isolated, and it identified quantum chromodynamics, the color gauge theory, as the correct theory of the strong interaction. The same period fixed the model's population: the tau lepton, discovered by Martin Perl at SLAC in 1976, and the bottom quark, found by Leon Lederman's team at Fermilab in 1977, completed the third generation. Thereafter the Standard Model was a finished framework awaiting confirmation, and confirmation arrived on schedule: the W and Z bosons at CERN in 1983, the top quark in 1995, the tau neutrino in 2000, and, on July 4, 2012, the Higgs boson itself, with the properties the theory had assigned it forty-eight years earlier. The chain that began with Dirac's reluctant positron ended with a fully populated table of matter, every particle in place.

The flavor of the discovery era is worth preserving, because the model was built against real resistance. The quark hypothesis was received with open skepticism by much of the community, Gell-Mann himself hedging for a time over whether the constituents were physical or merely algebraic, and the SLAC experiments that settled the matter were designed, in part, to test the alternative parton pictures. The electroweak theory's authors faced the objection

that the Higgs mechanism was a contrivance, an accounting trick for masses that no one could otherwise explain, and the trick's own boson went unhunted by some laboratories because it was not considered a sure thing to exist. Asymptotic freedom arrived as a surprise even to its discoverers, who had set out to prove the opposite, that the color theory could not be the strong force, and found their calculation showing that it must be. The Standard Model's smooth, inevitable appearance in the textbooks is the retrospective illusion that every successful science wears, and the chapters that follow try to keep the actual texture visible: a model assembled from proposals that were each, at the time, contestable, and that were each, in the end, verified.

The discovery chapter's dates repay a final summary in one breath, because the compression itself is the drama: 1928 the Dirac equation, 1932 the positron, 1954 Yang and Mills, 1957 parity's failure, 1961 Glashow's electroweak structure, 1964 the quarks, the color charge, and the Higgs mechanism, 1967 the Weinberg-Salam completion, 1971 renormalizability proved, 1973 the neutral current and asymptotic freedom, 1974 charm, 1976 the tau, 1977 the bottom, 1983 the W and Z, 1995 the top, 2000 the tau neutrino, 2012 the Higgs. Fifty-five years of construction, ending within a human lifetime of its beginning, with several of the constructors, Higgs most famously, alive at the finish. Science has few creations whose complete genealogy is so short and so witnessed, and the reader should hold the compression in mind while the rest of the book unpacks it: the Standard Model was assembled in the space of one working life, by people who mostly met each other.